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REVIEWER ACTIVITIES
ARMY-AR,MI
AIR FORCE-84
DLA-IS

USER ACTIVITIES:
ARMY-AV

P.A. OS
Other Cust AT 99
INTERNATIONAL INTEREST
PROCUREMENT SPECIFICATION MIL-B-5687
SUPERSEDES:

DD FORM 672-1 COORDINATED:

TITLE
BEARING,SLEEVE,PLAIN,SINTERED
BRONZE,OIL IMPREGNATED

MILITARY STANDARD
MS 17795
SHEET 1 OF 4

3120-0591

FED. SUP CLASS 3120

Ⓐ ENTIRE STANDARD REVISED

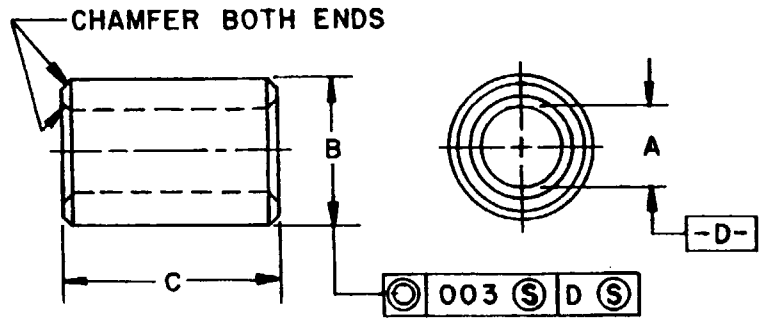


TABLE I. DIMENSIONS AND DASH NUMBERS.

NOMINAL ID	1/8			5/32		3/16			1/4			5/16			3/8			7/16	
A-BASIC ID	0.127			0.158		0.1895			0.252			0.3145			0.377			0.439	
B-BASIC OD	0.1905		0.253	0.253		0.253		0.3155	0.378		0.4405	0.4405		0.503	0.503		0.628	0.565	
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS
3/32	1	97	—	—	—	—	—	—	—	—	—								
1/8	2	129	5	214	161	9	193	—	—	—	—								
3/16	3	194	6	215	242	10	290	15	21	386	28								
1/4	4	258	7	216	322	11	387	16	22	516	29	36	645	243	43	773	52	—	—
5/16	207	323	8	217	403	12	483	17	23	645	30	37	806	244	44	967	53	—	—
3/8	208	387	211	218	483	13	580	18	24	773	31	38	967	245	45	1160	54	62	1354
7/16	209	451	212	219	564	14	677	19	25	902	32	39	1128	246	46	1354	55	63	1579
1/2	210	516	213	220	645	221	774	20	26	1031	33	40	1289	247	47	1547	56	64	1805
5/8	—	—	—	—	—	222	967	224	27	1289	34	41	1611	248	48	1934	57	65	2256
3/4						223	1160	225	228	1547	35	42	1934	249	49	2320	58	66	2707
7/8						—	1354	226	229	1805	233	238	2256	250	50	2707	59	67	3158
1						—	1546	227	230	2063	234	239	2578	251	51	3094	60	68	3609
1-1/8									231	2320	235	240	2900	252	255	3480	258	259	4061
1-1/4									232	2578	236	241	3223	253	256	3867	61	69	4512
1-1/2									—	3094	237	242	3867	254	257	4641	—	260	5414

APPROVED 14 FEB 1962 REVISED Ⓐ 9 November 1984

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REVIEWER ACTIVITIES
ARMY - AR,MI
AIR FORCE - 84
DLA - IS

USER ACTIVITIES:
ARMY - AV

TABLE I. DIMENSIONS AND DASH NUMBERS. -CONTINUED

NOMINAL ID	1/2			9/16		5/8			3/4			7/8		1					1-1/8	
A-BASIC ID	0.502			0.565		0.627			0.752			0.877		1.003					1.128	
B-BASIC OD	0.628		0.753	0.695		0.753		0.879	0.879		1.004	1.004		1.129		1.254		1.379	1.378	
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS
3/8	70	1547	78	—	—	—	—	—	—	—	—									
7/16	71	1805	—	—	—	—	—	—	—	—	—									
1/2	72	2063	79	86	2320	93	2578	100	108	3094	116									
5/8	73	2578	80	87	2900	94	3223	101	109	3867	117	125	4512	—	—	—	—	—		
3/4	74	3094	81	88	3480	95	3867	102	110	4640	118	126	5414	132	6188	139	6188	291		
7/8	75	3609	82	89	4061	96	4518	103	111	5414	119	127	6316	133	7219	140	7219	292		
1	76	4125	83	90	4641	97	5156	104	112	6188	120	128	7219	134	8250	141	8250	293	148	9281
1-1/8	261	4641	264	268	5221	269	5801	270	274	6961	276	281	8121	283	9281	286	9281	294	304	10441
1-1/4	77	5156	84	91	5801	98	6445	105	113	7734	121	129	9023	135	10313	142	10313	295	149	11602
1-1/2	262	6188	85	92	6961	99	7734	106	114	9281	122	130	10828	136	12375	143	12375	296	150	13922
1-5/8	263	6703	265	—	—	—	8379	271	275	10055	277	282	11730	284	13406	287	13406	297	305	15082
1-3/4	—	7219	266	—	—	—	9023	107	115	10828	123	131	12633	137	14438	144	14438	298	151	16242
1-7/8		8250	267				9668	272		11602	278			285	15469	288	15469	299	306	17402
2		—	—				10313	273		12375	124			138	16500	145	16500	300	152	18563
2-1/4		—	—				—	—		13922	279			—	18563	146	18563	301	153	20883
2-1/2										15469	280				20625	147	20625	302	154	23203
2-3/4										—	—				22681	289	22681	303	307	25523
3										—	—				24750	290	24750	—	308	27800

APPROVED 14 FEB 1962 REVISED (A) FOR CHANGES SEE PAGE 1

DD FORM 672-1 (COORDINATED)

PROCUREMENT SPECIFICATION MIL-B-5687

SUPERSEDES:

INTERNATIONAL INTEREST

P. A. OS Other Cust AT 99

TITLE
BEARING, SLEEVE, PLAIN, SINTERED
BRONZE, OIL IMPREGNATED

MILITARY STANDARD
MS 17795

SHEET 2 OF 4

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ARMY - AR, MI
AIR FORCE - 84
DLA - IS

USER ACTIVITIES:
ARMY - AV

TABLE I. DIMENSIONS AND DASH NUMBERS. CONTINUED.

NOMINAL ID	1-1/4			1-3/8		1-1/2			1-3/4		2			2-1/4		2-1/2		3	
A-BASIC ID	1.2535			1.378		1.504			1.753		2.004			2.254		2.505		3.006	
B-BASIC OD	1.504		1.630	1.629		1.755		1.880	2.005		2.380		2.505	2.631		3.006		3.507	
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS
1	155	10313	312	164	11344	173	12375	330											
1-1/8	309	11602	313	324	12762	327	13922	331											
1-1/4	156	12891	314	165	14180	174	15469	332											
1-1/2	157	15469	315	166	17016	175	18563	333	182	21656	—	—	—						
1-5/8	310	16758	316	325	18434	328	20109	334	342	23461	—	—	—						
1-3/4	158	18047	317	167	19852	176	21656	335	183	25266	189	26875	345						
1-7/8	311	19336	318	326	21270	329	23203	336	343	27070	344	30938	346	—	—	—	—	—	—
2	159	20625	319	168	22688	177	24750	337	184	28875	190	33000	347	197	37125	202	41250	362	49500
2-1/4	160	23203	320	169	25523	178	27844	338	185	32484	191	37125	348	354	41766	358	46406	363	55688
2-1/2	161	25781	321	170	28359	179	30938	339	186	36094	192	41250	349	198	46406	203	51563	364	61875
2-3/4	162	28359	322	171	31195	180	34031	340	187	39703	193	45375	350	355	51047	359	56719	365	68063
3	163	30938	323	172	34031	181	37125	341	188	43313	194	49500	351	199	55688	204	61875	366	74250
3-1/4											—	—	—	356	60328	360	67031	367	80438
3-1/2											195	57750	352	200	64969	205	72188	368	86625
3-3/4											—	—	—	357	69609	361	77344	369	92813
4											196	66000	353	201	74250	206	82500	370	99000

REQUIREMENTS:

1. THESE BEARINGS ARE NOT INTENDED FOR REAMING ON ASSEMBLY.
2. MATERIAL: SINTERED, COPPER BASE, POWDERED METAL IN ACCORDANCE WITH MIL-B-5687. TYPE I, GRADE 1 OR 2. YIELD STRENGTH IN COMPRESSION, 11,000 PSI (MIN) FOR 0.1% PERMANENT SET.
3. HARDNESS: 30-70 HRH.
4. POROSITY: (OIL CONTENT BY VOLUME) 20% MINIMUM.
5. LUBRICATION: BEARINGS ARE TO BE IMPREGNATED 18% BY VOLUME WITH OIL CONFORMING TO MIL-L-17331.
6. DIMENSIONS: ALL DIMENSIONS ARE IN INCHES.
7. DIMENSIONAL TOLERANCES AND CHAMFERS. SEE TABLES II, III, IV, AND V.
8. LOAD RATINGS: SEE TABLE VI AND REQUIREMENT 9.
9. PERMISSIBLE LOAD: FOR SHAFT VELOCITIES ABOVE 200 FEET PER MINUTE, THE PERMISSIBLE LOAD IS FOUND FROM THE FOLLOWING:

$$P = \frac{50{,}000}{V} \quad \text{WHERE} \quad \begin{array}{l} P = \text{SAFE LOAD, PSI OF PROJECTED AREA (W x BEARING ID x LENGTH)} \\ V = \text{SHAFT VELOCITY IN FT/MIN} \end{array}$$

10. STATIC CAPACITY: SPECIFIC STATIC CAPACITY IS 75% OF THE MAXIMUM PURE RADIAL LOAD WHICH WILL PRODUCE A PERMANENT SET OF LESS THAN 0.1% IN BEARING AT ANY CONTACT AREA BETWEEN SHAFT AND BEARING AT ZERO VELOCITY. (NOTE: THE SPECIFIC STATIC CAPACITY GIVEN IS INTENDED AS A GUIDE IN SELECTING THE PROPER BEARING AND NO TESTS ARE REQUIRED TO DETERMINE THIS FACTOR FOR ACCEPTANCE OF BEARING.)
11. STATIC LOAD: THE MAXIMUM LOAD SHOULD NOT EXCEED 8,500 PSI OF PROJECTED BEARING AREA (LENGTH x INSIDE DIAMETER OF BEARING).
12. RADIAL CRUSHING STRENGTH: THE RADIAL CRUSHING STRENGTH OF CYLINDRICAL PARTS SHALL BE NOT LESS THAN THE VALUE DETERMINED FROM THE FOLLOWING;

$$S = \frac{Lt^2 x 22{,}500}{D-t} \quad \text{WHERE} \quad \begin{array}{l} S = \text{RADIAL CRUSHING STRENGTH IN POUNDS} \\ D = \text{OD OF BEARING IN INCHES} \\ t = \text{WALL THICKNESS IN INCHES} \\ L = \text{LENGTH OF BEARING IN INCHES} \end{array}$$

PROCUREMENT SPECIFICATION MIL-B-5687
SUPERSEDES:
P.A. OS Other Cust AT 99
INTERNATIONAL INTEREST
TITLE BEARING, SLEEVE, PLAIN, SINTERED BRONZE, OIL IMPREGNATED
MILITARY STANDARD
MS 17795
SHEET 3 OF 4
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DD FORM 672-1 (COORDINATED)
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ARMY-AR,MI
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DLA-IS

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ARMY-AV

TABLE II. DIMENSIONAL TOLERANCES.

DIMENSION	BEARING SIZE	TOLERANCE
INSIDE DIAMETER "A" AND OUTSIDE DIAMETER "B"	UP TO 1.510 ID INCL	+.000 -.001
	1.511 TO 2.510	+.0000 -.0015
	2.511 TO 3.010	+.000 -.002
	3.011 TO 3.507	+.000 -.003
LENGTH "C"	UP TO 1.495 INCL	±0.005
	1.496 TO 2.990	±0.0075
	2.991 TO 4.000	±0.010
CONCENTRICITY (◎) FULL INDICATOR MOVEMENT (FIM)	OUTSIDE DIAMETER (BASIC)	TOLERANCE
	UP TO 1.379 1.380 TO 2.005 2.006 TO 3.507	0.003 0.004 0.005

TABLE III. CHAMFERS.

RANGE	SIZE
WALL THICKNESS UP TO AND INCLUDING 3/32"	BREAK EDGES
WALL THICKNESS GREATER THAN 3/32" UP TO 3" OUTSIDE DIAMETER	1/64 INCH x45 DEGREES
ALL BEARINGS 3" OUTSIDE DIAMETER AND OVER	1/32 INCH x45 DEGREES

TABLE IV. SHAFT RUNNING CLEARANCE.

SHAFT SIZE	CLEARANCE MIN
UP TO 0.760	0.0005
0.761 TO 1.510	0.0010
1.511 TO 2.510	0.0015
OVER 2.510	0.0020

TABLE V. RECOMMENDED PRESS FITS.

OUTSIDE DIAMETER "B"	PRESS FITS	
	MAX	MIN
UP TO 0.760	0.003	0.001
0.760 TO 1.510	0.004	0.0015
1.511 TO 2.510	0.005	0.002
2.511 TO 3.010	0.006	0.002
OVER 3.010	0.007	0.002

TABLE VI. PERMISSIBLE LOAD RATINGS.

SHAFT VELOCITY (FEET PER MINUTE)	PERMISSIBLE LOAD (POUNDS PER SQUARE INCH)
SLOW AND INTERMITTENT	4000
25	2000
26 TO 50 INCL	1000
51 TO 100 INCL	500
101 TO 150 INCL	325
151 TO 200 INCL	250
OVER 200	SEE REQUIREMENT 8

REQUIREMENTS - CONTINUED:

13. PART NUMBER. THE BASIC MILITARY STANDARD (MS) PART NUMBER IS FOLLOWED BY A DASH NUMBER SELECTED FROM TABLE I.

EXAMPLE: MS17795-1
- -1 — DASH NUMBER FROM TABLE I
- MS17795 — BASIC MS NUMBER

NOTES:
1. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSALS, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
3. NOT RECOMMENDED FOR MILITARY AIRFRAME APPLICATIONS.

APPROVED 14 FEB 1962 REVISED (A) FOR CHANGES SEE PAGE 1

FED. SUP CLASS 3120

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MILITARY STANDARD

MS 17795

SHEET 4 OF 4

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